

1. What is Programming?

Answer:

Programming is the process of writing instructions that tell a computer how to perform specific tasks.

2. What is a Class?

Answer:

A class is a blueprint or template used to create objects.

Example

```
class Student
{
    String name;
}
```

3. What is an Object?

Answer:

An object is an instance of a class that contains data and behavior.

Example

```
Student s1 = new Student();
```

Here `s1` is an object.

4. Difference Between Class and Object

Class	Object
Blueprint	Real instance
Logical entity	Physical entity

Example:

- Class = Car
 - Object = BMW Car
-

5. What is OOP?

Answer:

OOP (Object-Oriented Programming) is a programming approach based on objects and classes.

Main concepts:

- Encapsulation
 - Inheritance
 - Polymorphism
 - Abstraction
-

6. What is Encapsulation?

Answer:

Encapsulation means binding data and methods into a single unit and restricting direct access to data.

Benefit:

- Data security
-

7. What is Inheritance?

Answer:

Inheritance allows one class to acquire properties and methods of another class.

Benefit:

- Code reusability

Example:

```
class Animal {}  
class Dog extends Animal {}
```

8. What is Polymorphism?

Answer:

Polymorphism means one action can have many forms.

Example:

A method can behave differently depending on inputs.

Benefit:

- Flexibility
-

9. What is Abstraction?

Answer:

Abstraction means showing only essential details while hiding implementation details.

Example:

Driving a car without knowing engine internals.

10. What is a Function?

Answer:

A function is a block of code designed to perform a specific task.

Example:

```
int add(int a,int b)
{
    return a+b;
}
```

11. What is Recursion?

Answer:

Recursion is a technique where a function calls itself until a stopping condition is reached.

Example:

Factorial calculation.

12. What is an Array?

Answer:

An array is a collection of elements of the same data type stored in contiguous memory locations.

Example:

```
int arr[]={10,20,30};
```

13. Difference Between Array and Linked List

Array	Linked List
Fixed size	Dynamic size
Fast access	Sequential access
Continuous memory	Non-continuous memory

14. What is Exception Handling?

Answer:

Exception handling is a mechanism used to handle runtime errors and prevent program crashes.

Example:

```
try{
}
catch(Exception e){
}
```

15. What is an API?

Answer:

API (Application Programming Interface) is a set of rules that allows different software applications to communicate with each other.

Example:

Weather app getting data from a weather server.

16. What is REST API?

Answer:

REST API is a web service that uses HTTP methods to exchange data between client and server.

Methods:

- GET
 - POST
 - PUT
 - DELETE
-

17. Difference Between GET and POST

GET	POST
Retrieves data	Sends data
Data visible in URL	Data hidden
Safe operation	Creates/updates data

18. What is Debugging?

Answer:

Debugging is the process of finding and fixing errors in a program.

19. What is a Bug?

Answer:

A bug is an error or defect in a software application that causes incorrect behavior.

20. GreyOrange Project-Based Question

Q: What debugging experience do you have?

Answer:

During my chat application and Laravel projects, I encountered issues related to authentication, database queries, and API integration. I used logs, error messages, and testing to identify the root cause and resolve the problems.

Priority 4: DBMS Theory

1. What is Normalization?

Answer:

Normalization is the process of organizing data in a database to reduce redundancy and improve data consistency.

2. Why is Normalization Important?

Answer:

It reduces duplicate data, improves data integrity, and makes database management easier.

3. What is 1NF (First Normal Form)?

Answer:

A table is in 1NF if each column contains only atomic (single) values and there are no repeating groups.

4. What is 2NF?

Answer:

A table is in 2NF if it is in 1NF and all non-key attributes are fully dependent on the primary key.

5. What is 3NF?

Answer:

A table is in 3NF if it is in 2NF and has no transitive dependency.

6. What are ACID Properties?

Answer:

ACID properties ensure reliable database transactions.

ACID:

- Atomicity
 - Consistency
 - Isolation
 - Durability
-

7. What is Atomicity?

Answer:

Either the entire transaction completes or none of it happens.

8. What is Consistency?

Answer:

The database remains in a valid state before and after a transaction.

9. What is Isolation?

Answer:

Multiple transactions should not interfere with each other.

10. What is Durability?

Answer:

Once a transaction is committed, it remains permanently stored.

11. What is a Transaction?

Answer:

A transaction is a sequence of operations performed as a single unit of work.

Example:

- Money transfer
- Order placement

Priority 5: Operating System

1. What is a Process?

Answer:

A process is a program that is currently running.

Example:

- Chrome
- VS Code
- MySQL

2. What is a Thread?

Answer:

A thread is the smallest unit of execution within a process.

3. Difference Between Process and Thread

Process	Thread
Independent execution	Part of process
More memory	Less memory

Slower creation	Faster creation
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Interview Answer:

A process is an independent running program, whereas a thread is a lightweight execution unit within a process.

4. What is Multitasking?

Answer:

Running multiple processes simultaneously.

Example:

- Chrome + Spotify + VS Code
-

5. What is Deadlock?

Answer:

Deadlock occurs when two or more processes wait indefinitely for resources held by each other.

6. What is CPU Scheduling?

Answer:

CPU scheduling determines which process gets CPU time.

7. What is Memory?

Answer:

Memory stores data and instructions needed by running programs.

8. What is RAM?

Answer:

RAM (Random Access Memory) is temporary memory used while programs are running.

9. What is Virtual Memory?

Answer:

Virtual memory uses disk space as an extension of RAM.

10. Why is Linux Preferred for Servers?

Answer:

Linux is secure, stable, efficient, and supports high-performance server environments.

Priority 6: Computer Networks

1. What is a Network?

Answer:

A network is a collection of connected devices that communicate with each other.

2. What is an IP Address?

Answer:

An IP address is a unique identifier assigned to a device on a network.

Example:

192.168.1.10

3. What is DNS?

Answer:

DNS converts domain names into IP addresses.

Example:

google.com → IP Address

4. What is HTTP?

Answer:

HTTP is a protocol used for communication between web browsers and servers.

5. What is HTTPS?

Answer:

HTTPS is the secure version of HTTP that uses encryption.

6. Difference Between HTTP and HTTPS

HTTP	HTTPS
Not secure	Secure

Port 80	Port 443
No encryption	Encrypted

7. What is TCP?

Answer:

TCP is a reliable communication protocol that ensures data is delivered correctly.

8. What is UDP?

Answer:

UDP is a faster communication protocol but does not guarantee delivery.

9. Difference Between TCP and UDP

TCP	UDP
Reliable	Faster
Connection-oriented	Connectionless
Error checking	Minimal error checking

10. What is Client-Server Architecture?

Answer:

A client sends requests and a server processes those requests and returns responses.

Example:

- Browser → Client

- Website Backend → Server
-

Priority 7: Support Engineer Scenarios

These are very important because GreyOrange is a customer-facing support role.

1. Customer Says Application is Down. What Will You Do?

Answer:

First, I will gather details about the issue. Then I will check application logs, server status, database connectivity, and system resources. After identifying the root cause, I will resolve the issue or escalate it if necessary.

2. Customer Says Application is Slow. What Will You Check?

Answer:

I will check CPU utilization, memory usage, database performance, application logs, and network connectivity to identify the bottleneck.

3. Customer is Angry. How Will You Handle It?

Answer:

I will remain calm, listen carefully, understand the issue, provide regular updates, and work towards resolving the problem professionally.

4. What is Root Cause Analysis (RCA)?

Answer:

RCA is the process of identifying the actual reason behind a problem instead of only fixing the symptoms.

5. What Will You Do If You Cannot Resolve an Issue?

Answer:

I will collect all relevant information, document my findings, and escalate the issue to the appropriate team while keeping the customer informed.

6. What is Escalation?

Answer:

Escalation is the process of forwarding an issue to a higher-level support or development team when it cannot be resolved at the current level.

7. What is a Ticket?

Answer:

A ticket is a record of a customer issue that contains details, priority, status, and resolution information.

8. Why Are Communication Skills Important for a Support Engineer?

Answer:

Support Engineers interact with customers regularly. Clear communication helps understand problems, provide updates, and maintain customer satisfaction.

9. What Will You Do Before Escalating an Issue?

Answer:

I will reproduce the issue, check logs, verify system resources, collect error messages, document findings, and then escalate with complete information.

10. Why Do You Want a Software Support Role?

Answer (for you):

I enjoy troubleshooting and solving technical problems. Through my chat application and Laravel projects, I gained experience in debugging, database management, and issue resolution, which aligns well with the Software Support Engineer role.

Final High-Frequency Questions for GreyOrange

If you have only one day left, memorize these:

Linux

- Linux vs Unix
- chmod
- chown
- grep
- tail
- free -m
- htop

- scp

SQL

- Primary Key
- Foreign Key
- SELECT
- WHERE
- LIKE
- INNER JOIN vs LEFT JOIN
- DELETE vs DROP vs TRUNCATE

CS Fundamentals

- Process vs Thread
- DBMS vs RDBMS
- ACID Properties
- HTTP vs HTTPS
- TCP vs UDP

Support

- Application Down Scenario
- Application Slow Scenario
- RCA
- Escalation
- Customer Handling

These are the questions most likely to appear in a GreyOrange Software Support campus interview.